

Classical Electron Orbits under Variable Initial Velocity Angle and Nucleic Charge - Rhea Zhu

Overview

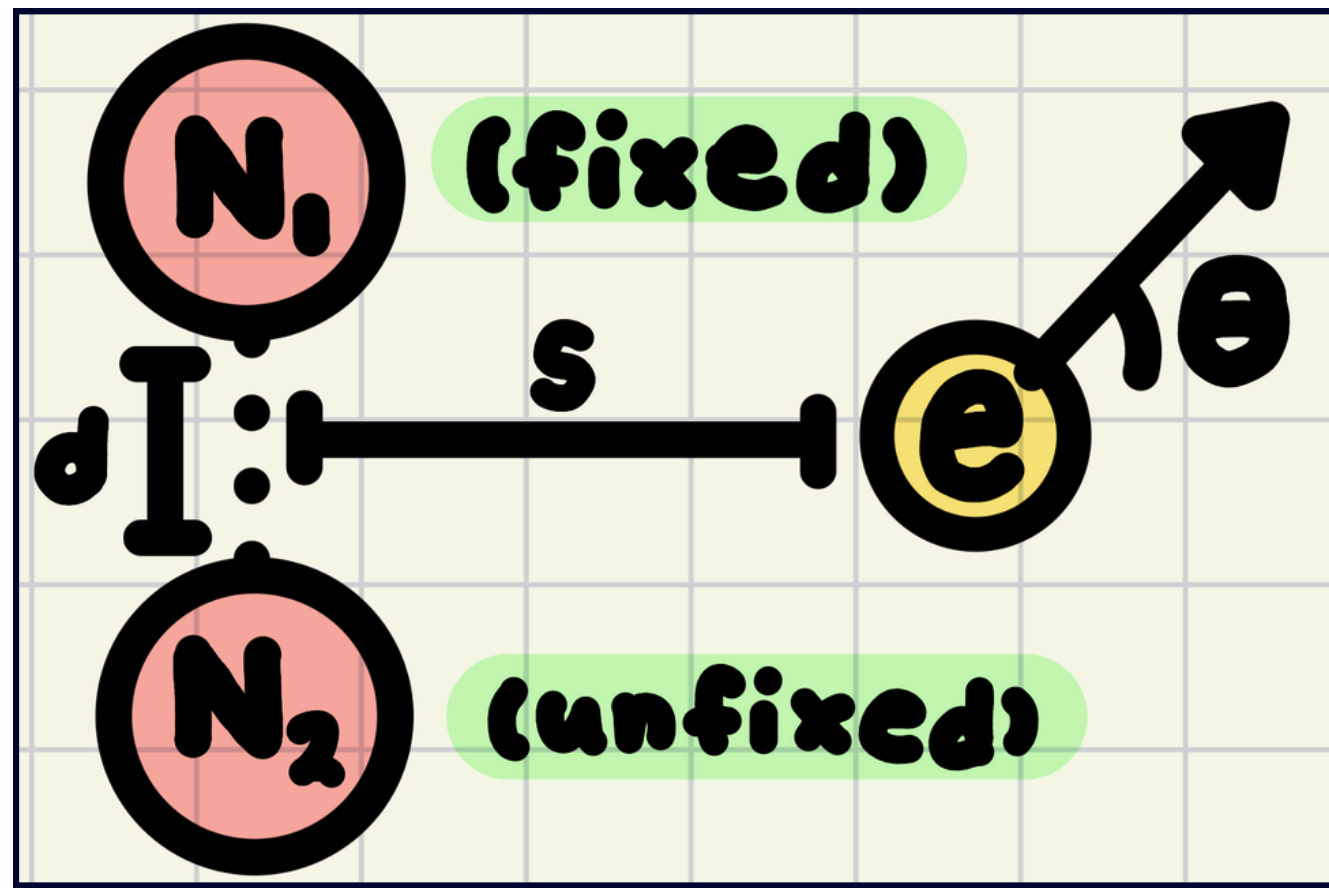


Fig 1. Diagram of simulation setup.

Electron orbits N_1 (fixed) and N_2 (variable) with initial velocity $2.18e16$ m/s (electron speed in hydrogen atom Bohr radius away) at an initial velocity angle θ .

$N_1 = -e$ | $s = a_0$ | $d = 1.2e-10$ m
Variables: N_2, θ

RQ1: What orbital paths are traced when varying θ and N_2 ?

RQ2: What combinations of θ and N_2 create stable orbital trajectories?

Defining Stability

1. Find distance **local minima**
2. **Index times** from distances
3. Calculate array of **periods**
4. Take **relative Std. Dev.**

$$\text{Stable: } \frac{dP}{P} \leq 0.1$$

I. Orbital Trajectories

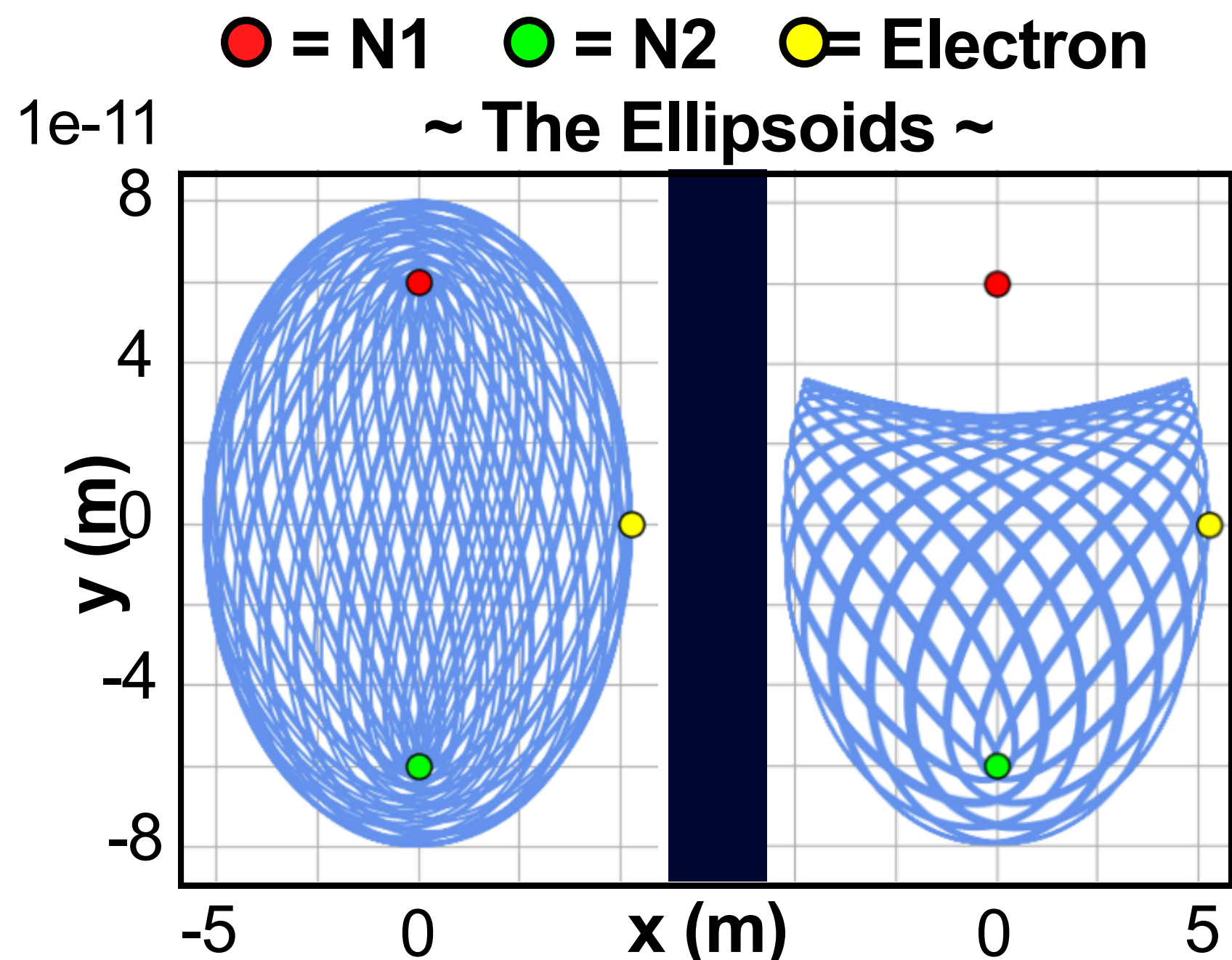


Fig 2. $N_2 = N_1$, (L); $N_2 = 5N_1$ (R), $\theta = 90$

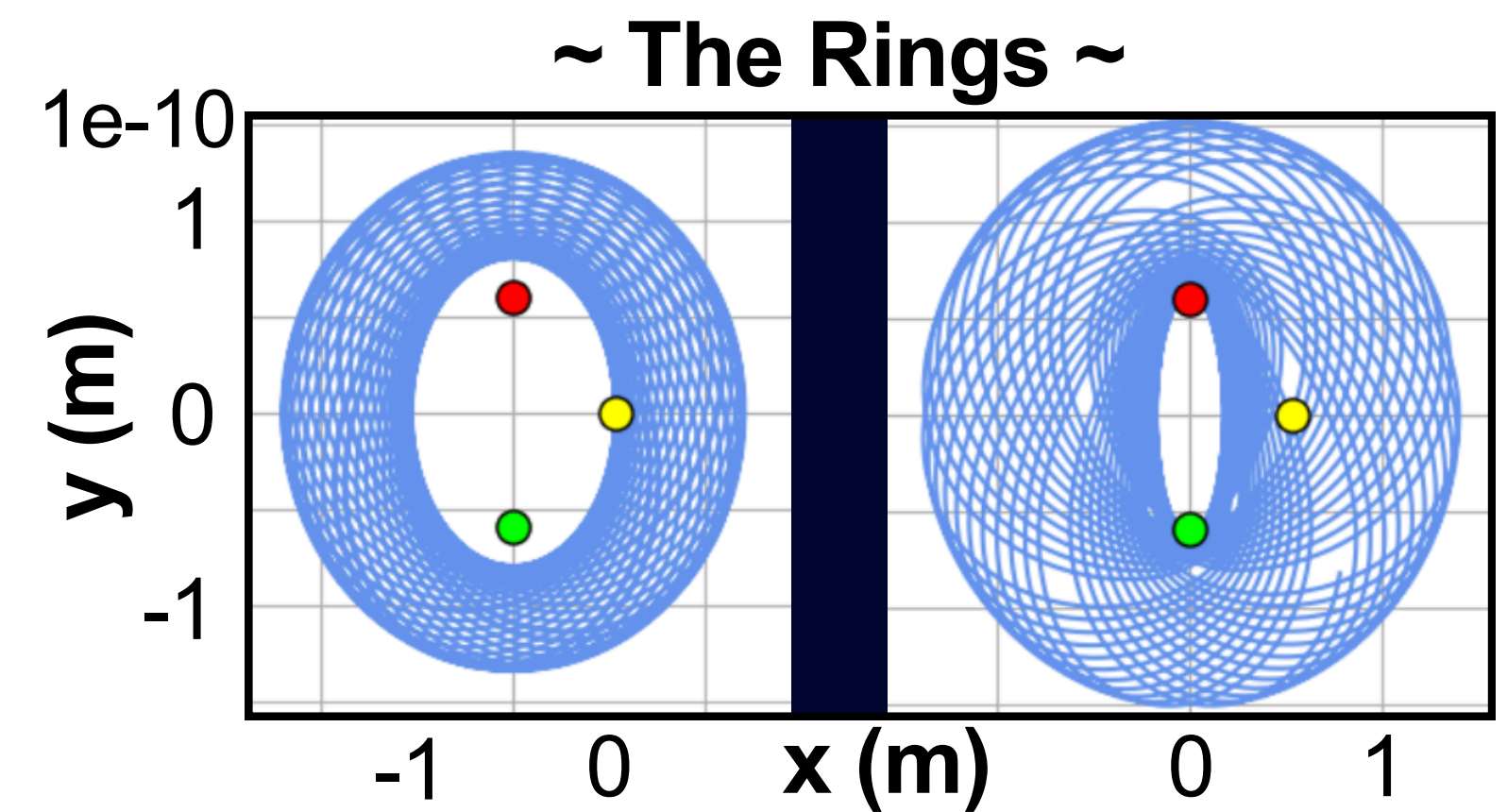


Fig 3. $N_2 = 0.2N_1$; $\theta = 90$ (L); $\theta = 70$ (R)

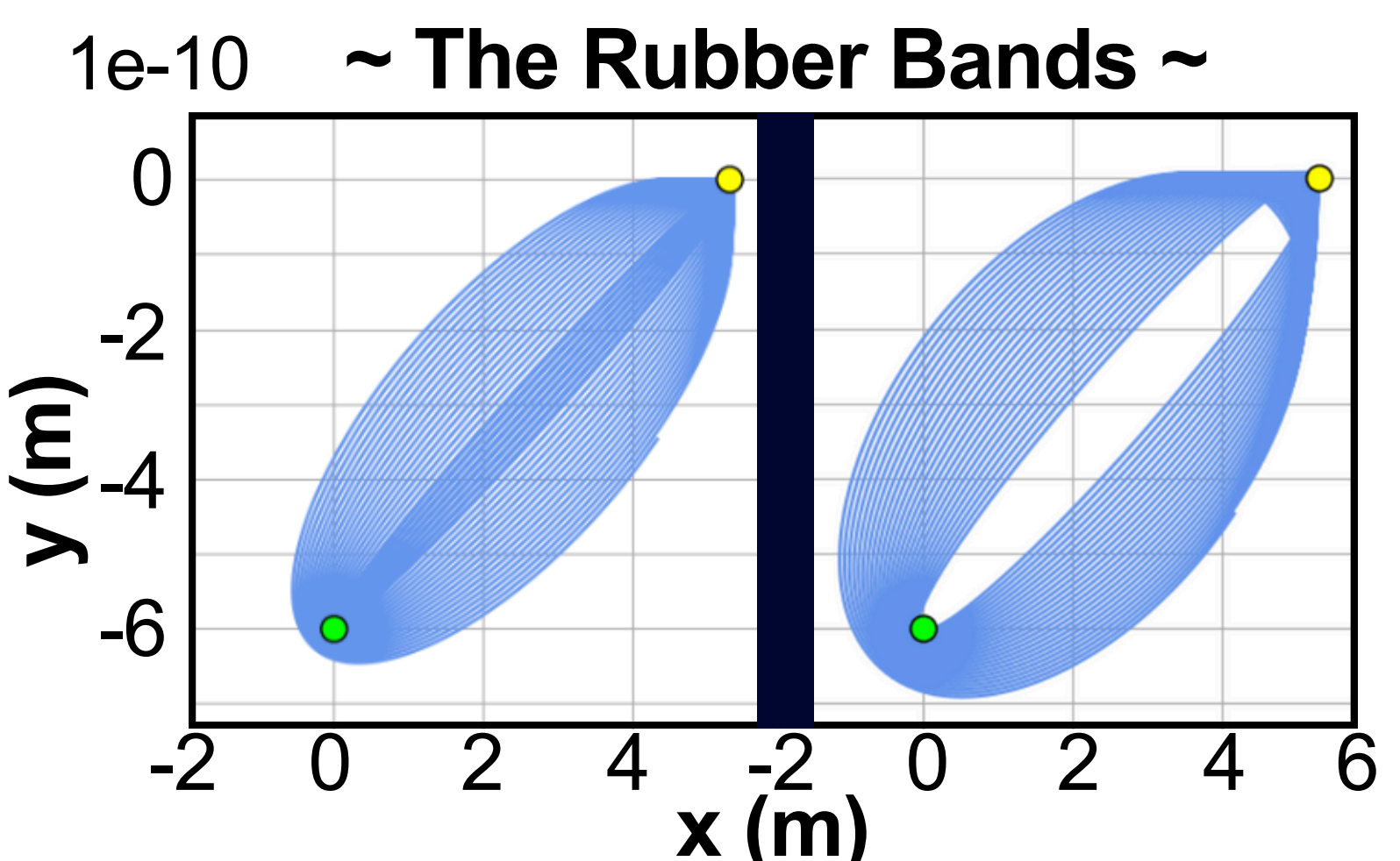


Fig 4. $N_2 = 100N_1$; $\theta = 20$ (L); $\theta = 90$ (R)

II. Stability Phase Space Scan

We run a **continuous scan** of 200 000 simulations.

Orbital Variability: $\theta \in [0^\circ, 90^\circ]$, $N_2 \in N_1 [0.01, 100]$

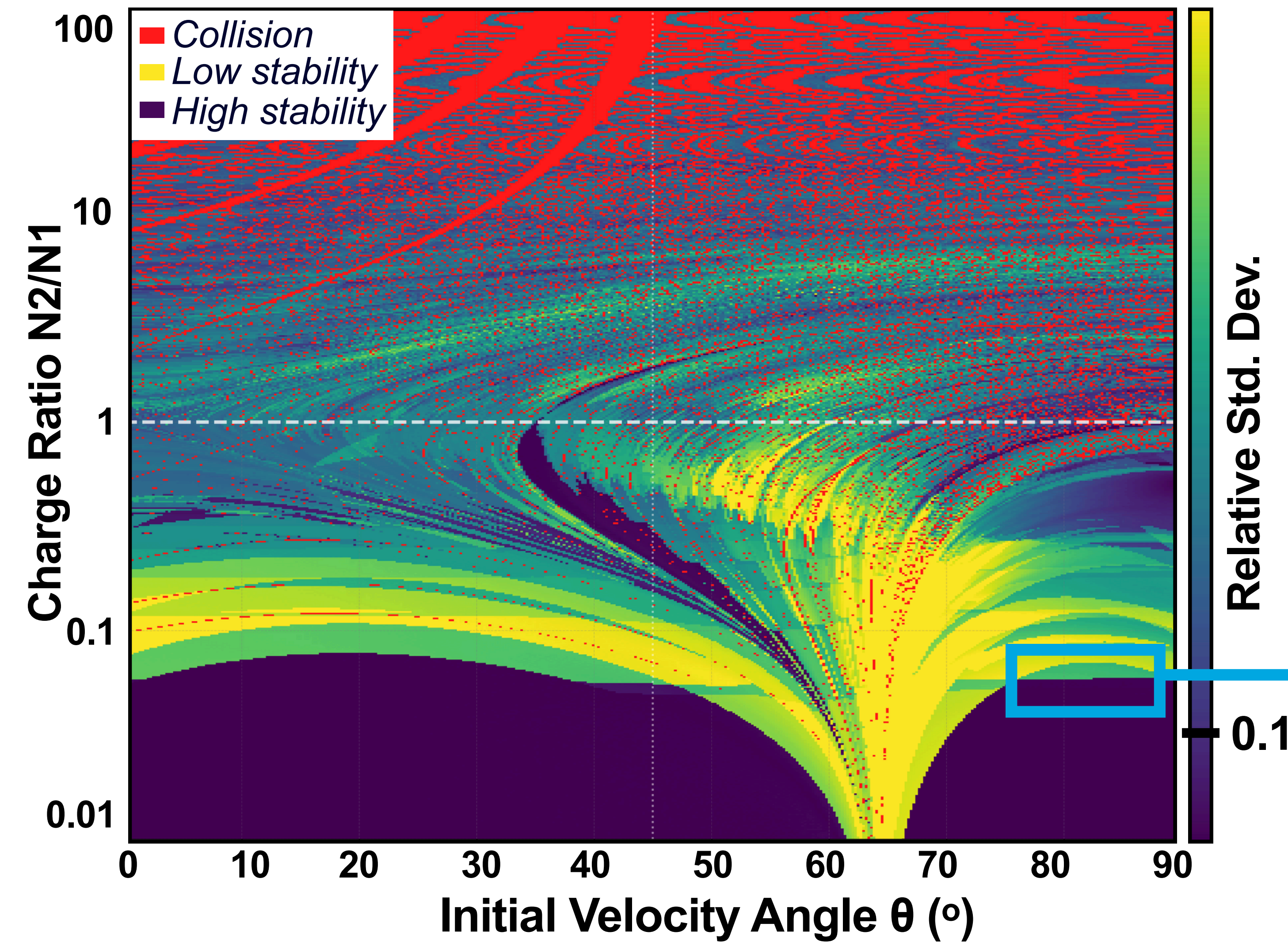


Fig 4. Continuous phase space scan of relative standard deviations over changing θ and N_2/N_1 .

Analyses & Patterns

- **Collisions** $\uparrow$ as $N_2/N_1 \uparrow$
- **Periodic & claw-mark collisions**
- **Neon bands** in lower third region
- **Curves** of periodicity / quasi-periodicity
- **Numerical artifacts**

III. Numerical Artifacts

We see an **abrupt shift from purple to green** not following a pattern. **Why?**

Histogram of Orbital Periods

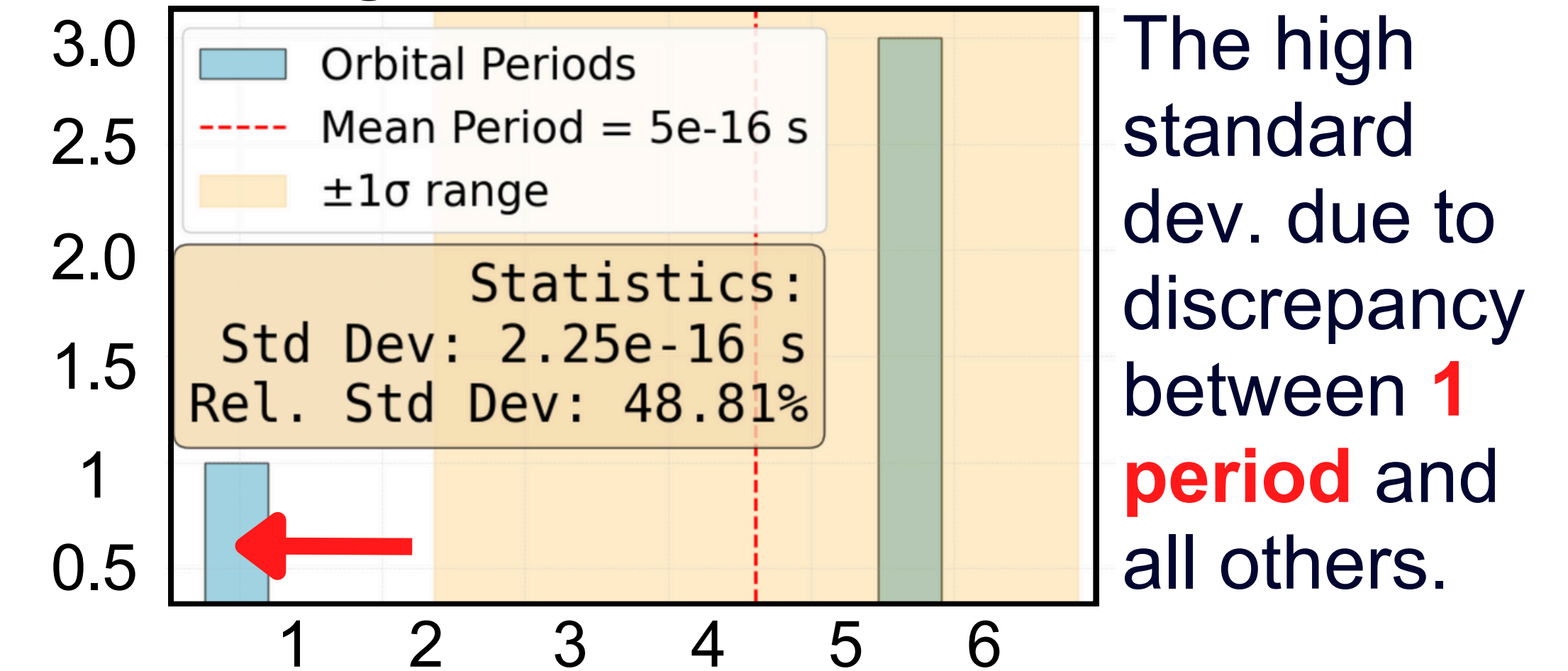


Fig 5. Periods when $N_2 = 0.02N_1$, $\theta = 5$.

This occurs from a **miscounting of periods** in the electron's last orbit, where **two detections of local minima are made**.

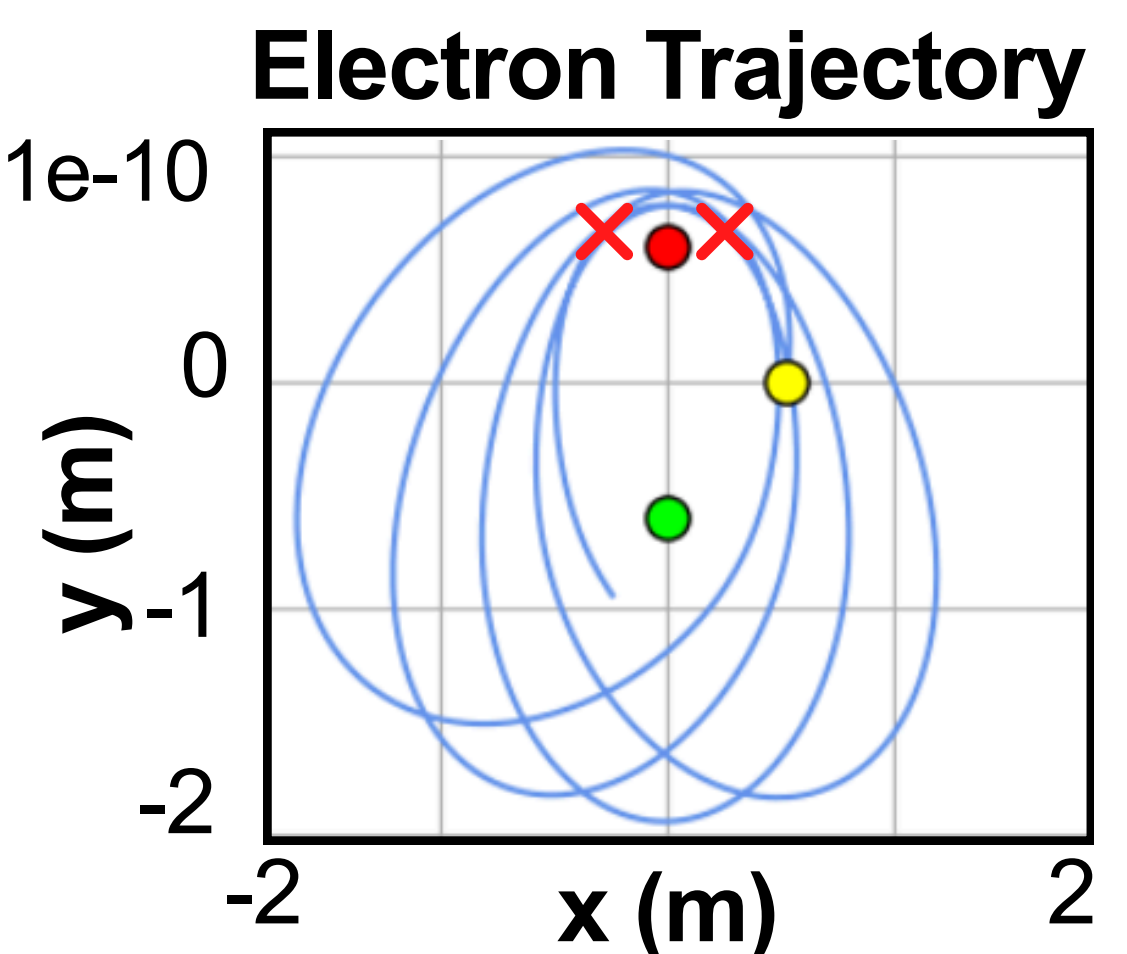


Fig 6. Electron path when $N_2 = 0.02N_1$, $\theta = 5$.

So, there exist limitations in our stability quantification, yielding numerical artifacts.

References

- [1] https://en.wikipedia.org/wiki/Coulomb%27s_law
- [2] https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve_ivp.html

Full Report:

